

Medvale Internal Medicine

Chapter 2E - Lung Cancer, Pulmonary Nodules, and Mediastinal Masses

Original study chapter for internal medicine learners. Source materials are used as coverage signals only; diagrams and teaching tables are original.

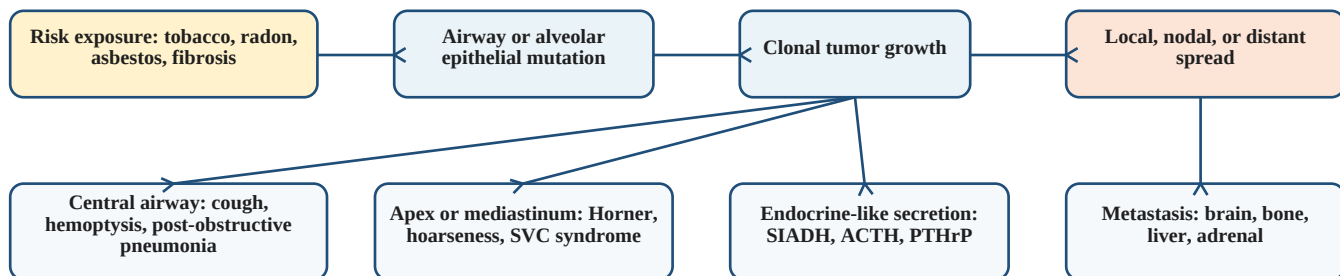
Coverage note

This segment follows the pulmonary sequence after pleural disease. It covers lung neoplasms, solitary pulmonary nodules, and mediastinal masses because these topics link obstructive symptoms, hemoptysis, recurrent pneumonia, imaging interpretation, tissue diagnosis, staging, and inpatient/ambulatory decision-making.

1. Why this matters clinically

Lung cancer is a high-yield internal medicine topic because it often presents indirectly. A patient may not arrive saying, "I have cancer." Instead, the first clue may be a chronic cough, mild hemoptysis, recurrent pneumonia in the same lobe, hoarseness, hypercalcemia, SIADH-related hyponatremia, venous distention from superior vena cava obstruction, or an incidental pulmonary nodule on CT. The clinician must connect symptoms to anatomy: a central endobronchial tumor causes obstruction and infection; an apical tumor invades the lower brachial plexus or sympathetic chain; mediastinal tumor burden compresses veins, nerves, airways, or the esophagus; and distant spread often involves brain, bone, adrenal glands, and liver.

The key internal medicine task is not to memorize every oncologic regimen. The key task is to recognize when lung cancer or a dangerous thoracic mass is plausible, determine the safest diagnostic pathway, distinguish surveillance from biopsy, stabilize complications, and refer early for multidisciplinary staging and treatment. In test questions, the correct answer often depends on whether the problem is a screening question, an incidental nodule question, a symptomatic mass question, or a known cancer staging question.



2. Core illness scripts

Problem	Typical clinical pattern	Most important next reasoning step
Central lung cancer	Cough, hemoptysis, wheeze, post-obstructive pneumonia, atelectasis, hilar mass, smoker or former smoker.	Localize airway involvement and obtain tissue by bronchoscopy or endobronchial ultrasound when accessible.
Peripheral lung adenocarcinoma	Incidental peripheral nodule or mass, possible pleural involvement; can occur in smokers and nonsmokers.	Risk-stratify nodule/mass and choose CT follow-up, PET-CT, biopsy, or surgical evaluation.
Small cell lung cancer	Central hilar/mediastinal mass, bulky nodes, early metastasis, SIADH or ectopic ACTH; heavy tobacco association.	Do not treat as surgically curable until staged; systemic therapy is central and urgent oncology referral is needed.
Pancoast tumor	Shoulder/arm pain, hand weakness, Horner syndrome from apical tumor invasion.	Recognize local invasion of C8-T2 and sympathetic chain; obtain imaging and tissue diagnosis.

Problem	Typical clinical pattern	Most important next reasoning step
Mediastinal mass	Chest pressure, cough, dyspnea, dysphagia, hoarseness, SVC syndrome, or incidental widened mediastinum.	Use compartment location on CT to narrow the differential and plan biopsy safely.

3. Risk factors and prevention anchors

Tobacco smoke remains the dominant risk factor for lung cancer, and risk rises with cumulative pack-years and shorter time since cessation. The relationship is biologically plausible: carcinogens injure airway and alveolar epithelium, repeated injury selects for clones with growth advantages, and additional inflammation promotes DNA damage. Smoking also synergizes with other exposures such as asbestos, meaning the combined risk is greater than either exposure alone. Radon exposure, prior thoracic radiation, pulmonary fibrosis or scarring, occupational exposures, and family history can also shift pretest probability upward.

Screening is different from diagnostic evaluation. Screening applies to asymptomatic high-risk adults. A symptomatic patient with hemoptysis, unexplained weight loss, persistent focal wheeze, or recurrent pneumonia does not need "screening"; that patient needs diagnostic imaging and evaluation. A useful numeric anchor is the USPSTF screening criterion: annual low-dose CT is recommended for adults 50 to 80 years old with at least a 20 pack-year smoking history who currently smoke or quit within the past 15 years, stopping when the quit interval exceeds 15 years or comorbidity makes curative treatment unrealistic.

Graph RAG relationship anchor

Smoking is linked to squamous cell carcinoma, small cell lung cancer, chronic bronchitis/COPD, recurrent infections, and cardiovascular disease. In a graph, tobacco exposure should connect to cancer risk, obstructive physiology, screening eligibility, perioperative risk, and prevention counseling.

4. Lung cancer categories and high-yield associations

The first major split is small cell lung cancer versus non-small cell lung cancer. Non-small cell lung cancer is a group that includes adenocarcinoma, squamous cell carcinoma, and large cell carcinoma. This distinction matters because staging, surgical candidacy, molecular testing, and systemic therapy differ. Small cell lung cancer is often centrally located, grows rapidly, spreads early, and behaves as a systemic disease at presentation. Non-small cell lung cancer can sometimes be treated with surgery or definitive radiation when localized, but metastatic disease requires molecular and immunologic characterization.

Cancer type	Typical location and risk pattern	Clues and complications	Management principle
Adenocarcinoma	Often peripheral; common overall; can occur in smokers and nonsmokers.	Peripheral nodule or mass, pleural involvement, malignant effusion; may be associated with scarring/fibrosis.	Stage with CT/PET and nodes; test advanced tumors for actionable alterations and PD-L1 when appropriate.
Squamous cell carcinoma	Often central; strong tobacco association.	Cavitation, endobronchial obstruction, hemoptysis, post-obstructive pneumonia; PTHrP-related hypercalcemia.	Bronchoscopy may obtain tissue if central; treat based on stage and resectability.
Large cell carcinoma	Often peripheral; diagnosis of exclusion when poorly differentiated NSCLC lacks glandular or squamous features.	Bulky mass, early metastasis possible, fewer classic exam clues.	Treat as NSCLC with stage-directed and molecularly informed therapy.
Small cell lung cancer	Often central/hilar; very strong tobacco association; bulky mediastinal nodes.	SIADH, ectopic ACTH, Lambert-Eaton syndrome, early brain/liver/bone/adrenal spread.	Systemic therapy is central; limited-stage may receive combined chemo-radiation; extensive disease is usually not surgical.

5. Presentation patterns: local, systemic, paraneoplastic, metastatic

Local symptoms reflect where the tumor sits. Airway invasion produces cough, wheeze, hemoptysis, atelectasis, or recurrent infection distal to obstruction. Pleural involvement causes pleuritic pain or malignant pleural effusion. Mediastinal involvement can compress the superior vena cava, recurrent laryngeal nerve, phrenic nerve, trachea, bronchi, or esophagus. A hoarse voice suggests recurrent laryngeal nerve involvement; an elevated hemidiaphragm suggests phrenic nerve palsy; dysphagia suggests esophageal compression; and venous distention of the face, neck, or upper extremities suggests superior vena cava syndrome.

Systemic cancer symptoms include anorexia, fatigue, unintentional weight loss, low-grade fevers, and night sweats. These symptoms are nonspecific, so they raise concern most strongly when paired with risk factors, focal chest findings, abnormal imaging, or laboratory abnormalities. Metastatic symptoms depend on site: brain metastases cause headache, seizure, focal neurologic deficits, or confusion; bone metastases cause focal pain or pathologic fracture; liver metastases can cause right upper quadrant discomfort or cholestatic laboratory abnormalities; adrenal metastases are often radiographic findings.

Syndrome	Classically linked tumor	Mechanism	Clinical clue
SIADH	Small cell lung cancer	Ectopic ADH effect	Euvolemic hyponatremia, low serum osmolality, inappropriately concentrated urine.
Ectopic ACTH	Small cell lung cancer	Tumor ACTH production	Hypokalemic metabolic alkalosis, hyperglycemia, weakness, rapid Cushingoid physiology.
Hypercalcemia	Squamous cell carcinoma	PTHrP production or bone metastasis	Confusion, constipation, polyuria, shortened QT; calcium often more urgent than the tumor workup.
Lambert-Eaton myasthenic syndrome	Small cell lung cancer	Autoantibodies against presynaptic voltage-gated calcium channels	Proximal weakness that may improve briefly with repeated use; autonomic symptoms.
Hypertrophic osteoarthropathy	NSCLC, especially adenocarcinoma	Periosteal reaction and clubbing-associated pathways	Digital clubbing, long bone pain, periostitis.

6. Diagnostic approach to suspected lung cancer

A chest radiograph may identify a mass, collapse, post-obstructive opacity, pleural effusion, or widened mediastinum, but a normal or nonspecific radiograph does not end the evaluation when clinical suspicion is high. Diagnostic CT of the chest with contrast is usually the central anatomic study because it evaluates mass size, relation to airways and vessels, pleural involvement, mediastinal lymphadenopathy, and potential biopsy routes. PET-CT adds metabolic information for staging and helps find occult nodal or distant disease, but inflammation and infection can be PET-avid, so PET positivity is not the same as histologic proof.

Tissue diagnosis is required before definitive treatment. The route depends on location and safety. Central endobronchial lesions are often approached by bronchoscopy. Mediastinal or hilar nodes may be sampled by endobronchial ultrasound-guided transbronchial needle aspiration. Peripheral lesions may require CT-guided transthoracic needle biopsy, navigational bronchoscopy, or surgical biopsy. In advanced disease, biopsy of the safest metastatic site may provide diagnosis and staging simultaneously. Before invasive biopsy, clinicians consider bleeding risk, anticoagulation, respiratory reserve, pneumothorax risk, and whether the chosen sample will be sufficient for histology and molecular testing.

Clinical situation	Preferred next step	Why
Symptomatic patient with suspicious CXR	CT chest with IV contrast, then tissue route planning.	Defines anatomy and helps avoid blind or low-yield biopsy.
Central visible airway lesion	Bronchoscopy with biopsy/washings; consider EBUS for nodes.	High yield and allows airway inspection.
Peripheral nodule/mass near chest wall	CT-guided biopsy or navigational bronchoscopy depending on location and pneumothorax risk.	Peripheral route may be higher yield but pneumothorax risk matters.
Bulky mediastinal nodes	EBUS-guided nodal sampling when accessible.	Can diagnose and stage in one procedure.
Known distant metastasis that is easy to sample	Biopsy metastatic site.	May confirm malignancy and stage IV disease with one procedure.

7. Staging and management principles

For non-small cell lung cancer, stage is commonly organized by tumor size/invasion, nodal spread, and metastasis. Earlier stages may be treated with surgery, stereotactic radiation, or combined modality therapy depending on operative candidacy and local spread. Locally advanced disease may need chemotherapy, immunotherapy, radiation, and sometimes surgery in carefully selected cases. Metastatic disease requires systemic therapy tailored by histology, performance status, molecular alterations, and immune markers.

Small cell lung cancer is often simplified clinically into limited-stage and extensive-stage disease. Limited-stage disease can be encompassed within a tolerable radiation field and may receive combined chemotherapy and radiation. Extensive-stage disease has spread beyond that boundary or includes distant metastasis and is treated primarily with systemic therapy. Although modern staging can use TNM language, the limited-versus-extensive framework remains a practical internal medicine anchor because it predicts whether thoracic radiation is being used with curative intent or systemic therapy is the main strategy.

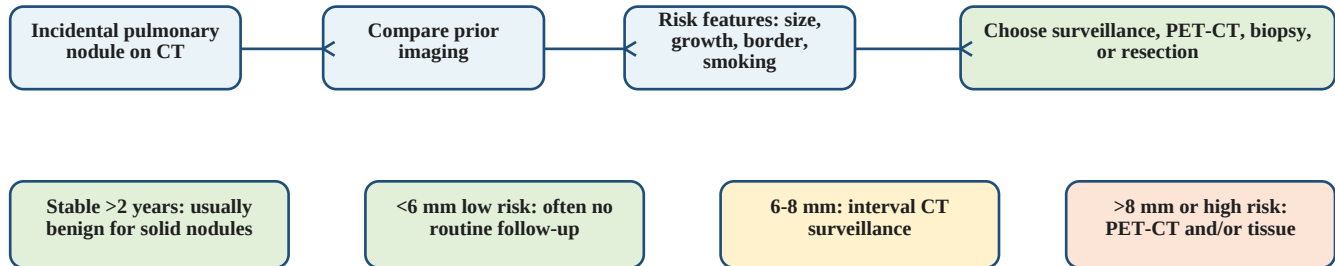
Do not miss

Treat emergencies before the complete cancer workup. Superior vena cava syndrome with airway or cerebral edema symptoms, massive hemoptysis, malignant pericardial tamponade, spinal cord compression, symptomatic brain metastasis, and severe hypercalcemia require urgent stabilization and specialist involvement.

8. Solitary pulmonary nodule: risk stratification before reflex biopsy

A pulmonary nodule is a rounded lung opacity, often found incidentally on CT. The immediate question is not simply "cancer or not cancer"; it is whether the probability of malignancy is low enough for surveillance, high enough for tissue diagnosis or resection, or intermediate enough for PET-CT and shared decision-making. The safest approach starts by comparing prior imaging. Stability over more than 2 years strongly favors benignity for many solid nodules, whereas growth over months raises concern for malignancy. Very rapid change over days is more suggestive of infection or inflammation than a typical lung cancer doubling time.

Nodule risk increases with age, smoking history, upper lobe location, spiculated or irregular margins, larger size, interval growth, emphysema, pulmonary fibrosis, and prior cancer history. Benign clues include small size, smooth margins, diffuse or central calcification, fat density suggesting hamartoma, and long-term stability. Subsolid nodules require special caution because ground-glass and part-solid nodules can grow slowly; persistent part-solid nodules are more concerning than transient inflammatory opacities.



Feature	Lower-risk direction	Higher-risk direction	Reasoning
Size	<6 mm often low risk in low-risk patients	>8 mm usually needs more active evaluation	Cancer probability rises as volume rises; volume doubles before diameter appears dramatically changed.
Growth	Stable >2 years for solid nodules	Clear growth on serial imaging	Growth indicates biologic activity; compare exact technique and slice thickness.
Border	Smooth, sharply circumscribed	Spiculated, lobulated, irregular	Spiculation reflects invasion or desmoplastic reaction.
Density	Benign calcification pattern or fat	Part-solid persistent nodule, eccentric calcification	Fat suggests hamartoma; part-solid persistence suggests adenocarcinoma spectrum.
Patient risk	Young, never smoker, no cancer history	Older age, heavy smoking, emphysema, prior malignancy	Pretest probability changes management threshold.

A practical size-based anchor from Fleischner-style incidental nodule management is: small solid nodules under 6 mm in low-risk patients often need no routine follow-up; 6-8 mm nodules commonly receive interval CT surveillance; and nodules over 8 mm generally require individualized consideration of CT at about 3 months, PET-CT, biopsy, or resection depending on risk and imaging appearance. These recommendations are for incidental nodules in adults and do not replace diagnostic workup for symptomatic patients, immunocompromised patients, or patients with known cancer.

Pitfall

Do not apply incidental nodule surveillance rules to a patient with hemoptysis, weight loss, post-obstructive pneumonia, or known extrathoracic cancer. Those patients have a different pretest probability and usually need diagnostic evaluation rather than routine surveillance.

9. Lung cancer mimics and differential diagnosis of a nodule or mass

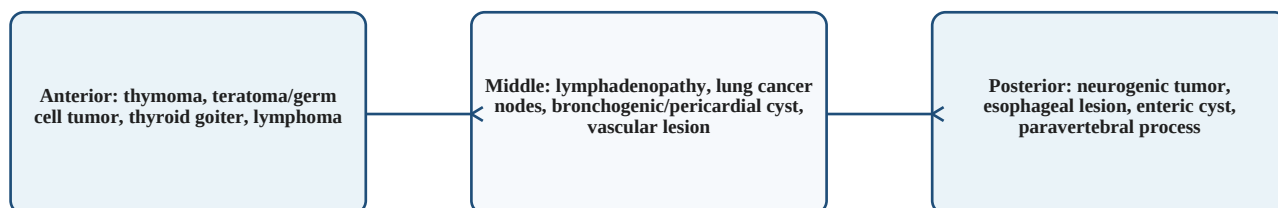
Not every mass-like opacity is cancer. Infectious granulomas from tuberculosis or endemic fungi, round pneumonia, septic emboli, rheumatoid nodules, granulomatosis with polyangiitis, sarcoidosis, pulmonary infarction, hamartoma, and organizing pneumonia can all mimic malignancy. The differential is shaped by immune status, geography, travel, occupational exposures, systemic symptoms, and imaging tempo. Fever, leukocytosis, and rapid radiographic change suggest infection, but malignancy can coexist with post-obstructive infection, so improvement with antibiotics should still be followed when the initial pattern is suspicious.

Mimic	Clue	How it differs from classic cancer logic
Granulomatous infection	TB or endemic fungal exposure; calcified nodes; systemic symptoms possible.	May be PET-avid and mimic cancer; microbiology or tissue may be needed.
Hamartoma	Fat density or popcorn-like calcification.	Benign tumor; often observed when imaging is classic.
Pulmonary infarct	Pleuritic pain, PE risk, wedge-shaped opacity.	Vascular event rather than epithelial tumor; evaluate for embolism.
Organizing pneumonia	Migratory opacities, autoimmune/drug/radiation context.	Inflammatory pattern; can present as mass-like consolidation.
Metastatic nodule	Known cancer, multiple nodules, variable sizes.	Primary lung cancer staging and biopsy logic differs from metastasis workup.

10. Mediastinal masses

Mediastinal masses are best organized by location because each compartment contains different structures. Anterior mediastinal masses classically include thymic tumors, teratoma or other germ cell tumors, thyroid lesions, and lymphoma. Middle mediastinal lesions often represent lymphadenopathy from lung cancer, lymphoma, sarcoidosis, infection, bronchogenic cysts, pericardial cysts, or vascular abnormalities. Posterior mediastinal masses are often neurogenic tumors, esophageal lesions, enteric cysts, or paravertebral processes.

Original location-based mediastinal mass map



The clinical presentation is usually caused by compression or invasion: cough and dyspnea from airway compression, dysphagia from esophageal compression, hoarseness from recurrent laryngeal nerve involvement, diaphragm elevation from phrenic nerve involvement, chest pain from invasion, and superior vena cava syndrome from venous obstruction. CT chest with IV contrast is the usual first high-yield imaging study because it defines the compartment, vascular relationships, cystic versus solid components, calcification, fat, and possible biopsy windows. Biopsy decisions should be deliberate: some vascular lesions, thyroid lesions, germ cell tumors, and thymic tumors have specific pre-biopsy considerations, and anterior mediastinal masses can create anesthesia risk if severe airway or vascular compression is present.

Compartment	High-yield differential	Clinical clue
Anterior	Thymoma, teratoma/germ cell tumor, thyroid goiter, lymphoma.	Myasthenia gravis with thymoma; young man with germ cell tumor; B symptoms with lymphoma.
Middle	Lymphadenopathy, lung cancer nodes, lymphoma, sarcoidosis, bronchogenic cyst, pericardial cyst, vascular lesion.	Hilar/mediastinal nodes, cough, SVC syndrome, or incidental widened mediastinum.
Posterior	Neurogenic tumor, esophageal mass, enteric cyst, paravertebral infection or hematoma.	Back pain, neurologic symptoms, dysphagia, or paraspinal findings.

11. Management of complications and inpatient decision-making

Internal medicine teams often meet patients during complications rather than at the moment of diagnosis. Superior vena cava syndrome can present with facial swelling, plethora, dyspnea, cough, headache, and distended chest wall veins. Mild chronic cases allow diagnostic confirmation before treatment, but severe airway compromise, laryngeal edema, cerebral edema, or hemodynamic instability demands urgent intervention with oncology, radiation oncology, interventional radiology, and critical care input. Malignant pleural effusion causes dyspnea and recurrent fluid accumulation; thoracentesis can both diagnose and palliate, while recurrent symptomatic effusions may need an indwelling pleural catheter or pleurodesis.

Massive hemoptysis is an airway emergency. The immediate threat is asphyxiation rather than exsanguination. The patient should be positioned with the bleeding lung down if known, large-bore access should be obtained, anticoagulants reversed when appropriate, and urgent bronchoscopy or bronchial artery embolization considered. Hypercalcemia from PTHrP or bone metastasis is treated with volume

repletion, antiresorptive therapy when appropriate, and treatment of the cancer. Brain metastases with edema, spinal cord compression, and pathologic fractures are oncologic emergencies that require urgent imaging and specialty involvement.

Complication	Immediate danger	First management priority
SVC syndrome	Airway/cerebral edema, venous congestion	Assess severity; urgent specialist treatment if airway, brain, or hemodynamic compromise.
Massive hemoptysis	Airway flooding/asphyxiation	Protect airway, bleeding side down if known, bronchoscopy/embolization pathway.
Malignant pleural effusion	Dyspnea, recurrent fluid	Thoracentesis for diagnosis and palliation; plan recurrence strategy.
Hypercalcemia	Arrhythmia, confusion, dehydration, renal injury	IV fluids, calcium-lowering therapy, evaluate PTHrP vs bone disease.
Spinal cord compression	Permanent neurologic deficit	Urgent MRI, corticosteroids when indicated, radiation/neurosurgical evaluation.

12. Differentials and mimics for exam reasoning

- Hemoptysis in a smoker:** lung cancer is important, but also consider bronchitis/COPD exacerbation, bronchiectasis, tuberculosis, pneumonia, pulmonary embolism, vasculitis, mitral stenosis, and anticoagulant-related bleeding.
- Recurrent pneumonia in the same lobe:** think obstruction until proven otherwise, especially from an endobronchial tumor, foreign body, or stricture.
- Hypercalcemia:** squamous cell lung cancer suggests PTHrP; widespread bony metastases suggest osteolysis; primary hyperparathyroidism remains common and should not be ignored.
- Hyponatremia:** small cell lung cancer can cause SIADH, but medications, adrenal insufficiency, hypothyroidism, heart failure, cirrhosis, and renal disease remain alternatives.
- Widened mediastinum:** consider lymphoma, thoracic aortic aneurysm/dissection, mediastinal hemorrhage, lymphadenopathy, goiter, thymic tumor, or technical/AP film artifact.

13. High-yield exam clues

Clue	Most likely association
Central mass + hypercalcemia	Squamous cell carcinoma with PTHrP.
Central mass + hyponatremia or ectopic ACTH	Small cell lung cancer.
Apical mass + shoulder pain + Horner syndrome	Pancoast tumor.
Hoarseness + mediastinal mass	Recurrent laryngeal nerve involvement.
Face/arm swelling + distended chest veins	Superior vena cava syndrome.
Peripheral lesion in nonsmoker	Adenocarcinoma is a common possibility, though tissue diagnosis is required.
Nodule stable >2 years on prior imaging	Usually benign if solid and technique is comparable.
Anterior mediastinal mass + myasthenia gravis	Thymoma.

14. Common pitfalls

- Calling a symptomatic patient a screening case. Screening is for asymptomatic high-risk patients; symptoms require diagnostic evaluation.
- Biopsying every nodule reflexively. Low-risk small nodules can often be followed; high-risk nodules need a deliberate pathway that balances cancer probability and biopsy risk.
- Assuming PET positivity proves cancer. Infection, inflammation, sarcoidosis, and post-treatment changes can be metabolically active.
- Forgetting that biopsy route depends on location. Central lesions, peripheral lesions, mediastinal nodes, and distant metastases have different safest sampling approaches.
- Missing emergencies while waiting for staging. Airway compromise, massive hemoptysis, spinal cord compression, symptomatic brain metastases, tamponade, and severe hypercalcemia are immediate problems.

15. Practice questions

1. A 67-year-old man with a 50 pack-year history has cough, hemoptysis, and a central cavitory mass. Serum calcium is 13.1 mg/dL. What is the likely histology and mechanism?

Answer: Squamous cell carcinoma with PTHrP-mediated hypercalcemia. The central location, cavitation, smoking association, and hypercalcemia fit this pattern.

2. A 64-year-old heavy smoker has a hilar mass, sodium 123 mEq/L, low serum osmolality, and inappropriately concentrated urine. What syndrome connects the laboratory abnormality to the lung finding?

Answer: SIADH from small cell lung cancer. The euvolemic hyponatremia pattern should trigger the paraneoplastic association.

3. A 59-year-old patient has an incidental 4 mm solid nodule on CT, no smoking history, and no prior cancer. What is the general management concept?

Answer: Low-risk small incidental nodules often do not require routine follow-up. The answer should acknowledge that exact follow-up depends on guideline context and radiologist assessment.

4. A patient has a right apical lung mass, right ptosis, miosis, anhidrosis, and severe shoulder pain radiating down the arm. What structure is involved?

Answer: A Pancoast tumor invading the sympathetic chain and lower brachial plexus/C8-T2 region.

5. A young adult has an anterior mediastinal mass and fatigable ptosis. What diagnosis should be considered?

Answer: Thymoma associated with myasthenia gravis.

References and source-use note

This chapter is original Medvale writing. Uploaded pulmonary material was used only to identify expected topic coverage, especially lung neoplasms, solitary pulmonary nodules, mediastinal masses, pleural disease, and pneumothorax sequencing. No source images or tables were reproduced.

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